

ECON345 Homework 9: Regression II and III

Due: November 5th, 11:59 PM

Develop your code in an R script first if that helps you think, then answer the questions (including your code) in an R Markdown file. Comment your code to explain what you're doing — partial credit rewards a reasonable, well-documented attempt, not just a correct final answer. For any part you can't produce output for, briefly explain what you tried and what went wrong.

Throughout this homework, use `broom::tidy()` and `broom::glance()` to read model output — not `summary(model)$coefficients` and not `$` on a data frame.

Data

These questions use `redfin_clean.csv`, home sales in DC, Maryland, and Virginia, filtered to `ppsf < 2000` to remove a few outliers. Key variables: `ppsf` (sale price per square foot, USD), `metro_distance` (miles to the nearest metro station), `state`, `property_type`, `beds`, `baths`, and `square_feet`.

For Questions 1 and 2, use `redfin_houses`, which further restricts the data to the three most common property types:

```
redfin_houses <- redfin_clean %>%  
  filter(property_type %in% c("Condo/Co-op", "Single Family Residential",  
                             "Townhouse"))
```

Question 1: Indicator Variables (20 points)

Question 1.1

Using `redfin_houses`, fit a regression of `ppsf` on `metro_distance` and `property_type`. Use `broom::tidy()` to display the results.

Question 1.2

Which `property_type` is the base group? How can you tell from the output?

Question 1.3

Verify that your code produces the output below (if you can't reproduce it, you can still answer 1.4 and 1.5):

term	estimate	std.error	statistic	p.value
(Intercept)	508.747	8.628	58.965	0.000
metro_distance	-84.500	5.761	-14.667	0.000
property_typeSingle Family Residential	-28.548	11.671	-2.446	0.015
property_typeTownhouse	-25.870	11.181	-2.314	0.021

Question 1.4

Pick one `property_type` indicator. Is it statistically significant at $\alpha = 0.05$? State the comparison it's measuring (which two groups, holding what fixed).

Question 1.5

The average home in `redfin_houses` is about 1591 square feet (computed with `pull()`, not `$`). Using the indicator you chose in 1.4, is its effect economically significant for a home of that size? Show the dollar amount you're basing that on.

Question 2: Interaction Terms (20 points)

Question 2.1

Using `redfin_houses`, fit a regression of `ppsf` on `metro_distance`, `property_type`, and the interaction between them.

Question 2.2

Verify that your code produces the output below (if not, you can still answer 2.3–2.5):

term	estimate	std.error	statistic	p.value
(Intercept)	535.598	9.743	54.970	0.000
metro_distance	-118.365	8.333	-14.204	0.000
property_typeSingle Family Residential	-142.339	19.370	-7.349	0.000
property_typeTownhouse	-46.804	15.811	-2.960	0.003
metro_distance:property_typeSingle Family Residential	99.638	13.772	7.235	0.000
metro_distance:property_typeTownhouse	27.079	13.552	1.998	0.046

Question 2.3

Using `broom::tidy()` and `pull()`, compute the slope on `metro_distance` for the base property type, and separately for one other property type (base slope plus that type's interaction coefficient).

Question 2.4

Compare `broom::glance()` for this model to your model from Question 1. Which has the higher R^2 ?

Question 2.5

Does a higher R^2 by itself tell you the interaction model is “better”? What else would you want to check before preferring it?

Question 3: Regression Specification (30 points)

For each equation below, write the `lm()` code that would specify it (assume all variables come from `redfin_clean`), and interpret the named coefficient in your own words – direction, size, and units, holding everything else in the equation fixed.

Question 3.1 (10 points)

$$\widehat{\text{ppsf}} = 300 - 50 \times \text{metro_distance} + 0.05 \times \text{square_feet}$$

Interpret the coefficient on `square_feet`.

Question 3.2 (10 points)

$$\widehat{\text{ppsf}} = 250 - 60 \times \text{metro_distance} + 10 \times \text{beds} + 5 \times (\text{beds} \times \text{metro_distance})$$

Interpret the interaction coefficient (the one multiplying `beds` and `metro_distance`).

Question 3.3 (10 points)

$$\widehat{\text{ppsf}} = 400 - 45 \times \log(\text{metro_distance})$$

Interpret the coefficient on `log(metro_distance)`. Be careful: this is not the same calculation as 3.1 or 3.2 – say what a 1% increase in `metro_distance` predicts for `ppsf`.

Question 4: Choosing a Model (10 points)

Question 4.1

Fit three regressions of `ppsf` on `metro_distance` using `redfin_clean`: (a) no grouping variable, (b) adding `state` as an indicator, (c) adding `state` as an interaction with `metro_distance`. Compare their R^2 with `broom::glance()`.

Question 4.2

In a few sentences, say which of the three you’d report as your main result if this were your class project, and why – reference both statistical and economic significance, not just R^2 .

Submission

Upload two files to Canvas before the deadline:

- your `week_09.Rmd`, and
- the **knitted PDF**.

As always: comment your code, and if you get stuck, show your work. If your `.Rmd` won't knit, get each piece working before you move on to the next, and restart R and knit clean if results look stale.